Risk Premium Inference from (Noisy) Option Price Panels

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Brief Description of Research Idea

Compare the finite sample performance of implied state vs. particle filtering when conducting joint $\mathbb{P}$-$\mathbb{Q}$ estimation on noisy option price panels focusing on the implications this has for recovered (equity) risk premium components and their error distributions/confidence bands.

Tentative Outline of Building Blocks

1. Implement an efficient simulation scheme for a Heston (Heston, 1993) state space simulation and associated option price panels.
2. Implement IS estimation à la Boswijk et al. (2015).
3. Implement particle filtering à la Hurn et al. (2015).
4. No noise/best case.
5. Noisy option panel case with an assorted menu of assumptions regarding option price error distributions.

Rationale

A. Noisy option data is something I had to deal with both for the first two papers, outside academia and having spent some more time looking at post 2018 academic work it seems that this is an emerging topic with a few papers that touch on this in a different context: Andersen et al. (2021), Duarte et al. (2024).

B. Topic itself ties in well with the other two papers in my thesis.

C. I have sufficient background to work on this independently.

1 Assumed Data Generating Proces And Option Pricing

We consider a filtered probability space $\left(\Omega, \mathcal{F}, \{\mathcal{F}\}_{t\geq 0}, \mathbb{P}\right)$ satisfying the usual conditions[1]. Under the historical probability measure, $\mathbb{P}$, the dynamics of the log-price process $\log(S_t)$ and of the stochastic volatility process V_t, which make up the state vector $X_t = [\log(S_t), V_t]$, are assumed to be described by the pair of s.d.e.:

$$\mathrm{d}\log(S_t) = \Big((r_t - q_t) + \big(\eta - 1/2\big)\, V_t\Big)\, \mathrm{d}t + \sqrt{V_t}\, \mathrm{d}W^{\mathbb{P}}_{t,1}; \tag{1.1}$$

$$\mathrm{d}V_t = k(\bar{V} - V_t)\, \mathrm{d}t + \sigma_v \sqrt{V_t} \left(\rho\, \mathrm{d}W^{\mathbb{P}}_{t,1} + \sqrt{1-\rho^2}\, \mathrm{d}W^{\mathbb{P}}_{t,2}\right). \tag{1.2}$$

$\left(\mathrm{d}W^{\mathbb{P}}_{t,1}, \mathrm{d}W^{\mathbb{P}}_{t,2}\right)$ are $\mathcal{F}_t$-adapted Brownian motion processes. $\{r\}_{t\geq 0}$ and $\{q\}_{t\geq 0}$ are $\mathcal{F}_t$-adapted processes (e.g., deterministic) which describe the risk-free rate of interest and the continuous dividend yield respectively. η is a parameter which governs the size of the equity risk premium. The local stochastic volatility process mean reverts at a mean reversion rate κ to the long run mean level $\bar{V}$. σ_v governs the volatility of volatility and ρ represents the instantaneous correlation between returns and volatility.

Under the risk neutral probability measure, $\mathbb{Q}$, the model dynamics are assumed to be:

$$\mathrm{d}\log(S_t) = \big((r_t - q_t) - 1/2 V_t\big)\, \mathrm{d}t + \sqrt{V_t}\, \mathrm{d}W^{\mathbb{Q}}_{t,1}; \tag{1.3}$$

$$\mathrm{d}V_t = k(\bar{V} - V_t)\, \mathrm{d}t + \eta_V V_t\, \mathrm{d}t + \sigma_v \sqrt{V_t} \left(\rho\, \mathrm{d}W^{\mathbb{Q}}_{t,1} + \sqrt{1-\rho^2}\, \mathrm{d}W^{\mathbb{Q}}_{t,2}\right). \tag{1.4}$$

$\left(\mathrm{d}W^{\mathbb{Q}}_{t,1}, \mathrm{d}W^{\mathbb{Q}}_{t,2}\right)$ are Brownian motions under the (equivalent) martingale pricing measure $\mathbb{Q}$.

Jointly, 1.3 and 1.4 are known as the Heston model, which features local stochastic volatility with a leverage effect while providing semi-closed form solutions for option prices. This model is a "staple" stochastic volatility model for equity index modeling. It is used in a large number of empirical studies, has a number of well documented limitations, some of which have been remedied by enriching the model specification through the addition of jumps and/or other (latent) stochastic states.

Given the focus of the study, attention is restricted to the simplistic set-up of the Heston model for a host of reasons. On the one hand, for most Heston model extensions, econometric approaches to parameter estimation which employ option price data come with increased computational costs. Since in the current set-up this model is known

[1] See e.g., Protter (2005), p. 3.

to be the data generating process, there's no potential misspecification issue at play. Secondly, extensions come with specific estimation challenges which typically require tailored estimation designs to obtain well-behaved parameter estimates therefore limiting the comparability of results obtained throughout the study.

1.1 Parameter Sets

Let θ denote the vector of parameters featured in the model. We assume $\theta \in \Theta$ where Θ is a compact set from the domain of valid data-generating process parameters. For the Heston model specification previously introduced, the value of θ corresponding to the data-generating process introduced in (1.1) - (1.4) is $\theta = [\eta, \kappa, \bar{v}, \sigma_v, \rho, \eta_v]$. Note that subsets of the parameter set play a role in either the $\mathbb{P}$-measure specification, the $\mathbb{Q}$-measure specification or in both. E.g., the Heston parameter set can be decomposed into 3 disjoint subsets: $\theta = \left[\theta^{\mathbb{P}}, \theta^{\mathbb{P}\cap\mathbb{Q}}, \theta^{\mathbb{Q}}\right]$, where $\theta^{\mathbb{P}} = \eta$, $\theta^{\mathbb{P}\cap\mathbb{Q}} = [\kappa, \bar{v}, \sigma_v, \rho]$, and $\theta^{\mathbb{Q}} = \eta_v$.

1.2 State Vector

Assuming we have a time series of state vector observations:[2] $X_1, \ldots, X_T$, observed at equally-spaced timepoints $t_1, \ldots, t_T$ with $t_{i+1} - t_i = \Delta > 0, \forall i \geq 1$. Denote by $f(X_{i+1}|X_i; \theta^{\mathbb{P}}, \theta^{\mathbb{P}\cap\mathbb{Q}})$ the transition density of the state vector X_t under the historical probability measure. Note that the transition density depends on $\theta^{\mathbb{P}}$ and $\theta^{\mathbb{P}\cap\mathbb{Q}}$ which represent the subset of parameters that describe the model dynamics under $\mathbb{P}$.

By using option price data the full set of parameters – including the additional ones that characterize model dynamics under the risk neutral pricing measure – can be estimated, i.e., inference on $\theta^{\mathbb{Q}}$ can be conducted while at the same time increasing sample identification potential for the $\theta^{\mathbb{P}\cap\mathbb{Q}}$ parameter set.

1.3 Option prices

Suppose that, in addition to time series observations of returns, synchronous observations of option price panel data and time series data on the underlying are available. Denote the option prices by $p_i(T, K)$ where T, K represent the maturity and the strike of the option contract. Option prices are commonly quoted in terms of their Black-Scholes implied volatility. Let $P(X_i, T, K; \theta^{\mathbb{P}\cap\mathbb{Q}}; \theta^{\mathbb{Q}})$ be the corresponding model price, e.g., the option price generated by the base model specification under the equivalent martingale

[2]In practice such a series never exists as the volatility, V_t, is an unobservable, i.e., latent state process

pricing measure for given state vector instance X_i and parameter set values $\theta^{\mathbb{P}\cap\mathbb{Q}}$ and $\theta^{\mathbb{Q}}$. If option prices were observed without error and the model is not misspecified, the observed option prices would coincide with their model-based counterparts. In practice the two prices differ. Typical modelling approaches assume away model misspecification and attribute the difference to measurement errors (e.g., microstructure noise). This assumption can in practice have consequences for the sample properties of the estimator of choice and thus warrants further consideration.

1.3.1 Option Prices with Observation Errors

Suppose that, at each observation date t_i, a panel of option price quotes is observed for a set of contracts from ranging from $j = 1$ to $j = q_i$ where $q_i \in \mathbb{N}^*$. This set is denoted by $\mathcal{C}_i = \{(T_i^{(j)}, K_i^{(j)})\}_{j=1}^{q_i}$ and the resulting price vector by $p_i^{\text{obs}} = \left(p_i^{(1),\text{obs}}, \ldots, p_i^{(q_i),\text{obs}}\right)^\top$, where $p_i^{(j),\text{obs}} = p_i^{\text{obs}}(T_i^{(j)}, K_i^{(j)})$ is the observed price for option contract j.

Similarly, let $p_i^0 = p_i^{\text{model}} = \left(p_i^{(1),0}, \ldots, p_i^{(q_i),0}\right)^\top$ be the vector of corresponding model-based prices implied by $P(X_i, T, K; \theta^{\mathbb{P}\cap\mathbb{Q}}; \theta^{\mathbb{Q}})$. Let $\sigma_i^0 = \left(\sigma_i^{(1),0}, \ldots, \sigma_i^{(q_i),0}\right)^\top$ denote the corresponding Black-Scholes implied volatility vector.

The difference between p_i^{obs} and p_i^{model} is interpreted as option observation error. This difference can be, inter alia, attributed to microstructure effects, stale quotes or price discreteness due to minimum tick sizing or quote rounding conventions. Duarte et al. (2024) suggests that such frictions can be quantitatively important for inference involving option-implied risk premia.

The proposed simulation design therefore does not commit ex-ante to a single parametric error law but instead considers three observed-panel specifications generated from the same clean model-implied panel. Each specification is meant to reflect empirical concerns raised by the recent literature while keeping the data generating exercise tractable. The three specifications are introduced in implied-volatility space. The corresponding contaminated prices are then obtained through the Black-Scholes pricing formula, rounded to the relevant tick size and projected back into the no-arbitrage price interval. This keeps the treatment of all three observation error specifications internally consistent while preserving both price and implied-volatility representations of the contaminated panel.

1.3.1.1 Design A: Low I.I.D. noise

First, a low-noise benchmark is introduced in which observation errors are independent across contracts, but the marginal scale of the error is allowed to vary with moneyness and maturity. This reflects the fact that option observation errors need not have the same relative impact across the surface and can enter the behaviour of option-based estimators through local features of the panel, in particular for contracts close to the spot or forward price as discussed in Chong and Todorov (2024). For $j = 1, \ldots, q_i$, let

$$m_i^{(j)} = \log \left(\frac{K_i^{(j)}}{F_i^{(j)}} \right), \qquad \tau_i^{(j)} = T_i^{(j)} - t_i,$$

where $F_i^{(j)}$ denotes the corresponding forward price. The marginal implied-volatility observation error scale is specified as

$$s_{\sigma,i}^{(j)} = s_\sigma(m_i^{(j)}, \tau_i^{(j)}) = \alpha_0 \left(1 + \alpha_m |m_i^{(j)}| + \frac{\alpha_\tau}{\sqrt{\max(\tau_i^{(j)}, \tau_{\min})}} \right). \tag{1.5}$$

The first specification is intended to mimic a liquid-market benchmark in which implied-volatility observation errors are independent across contracts conditional on the clean panel:

$$\tilde{\sigma}_i^{(j),A} = \max \left\{ \sigma_{\min}, \sigma_i^{(j),0} + s_{\sigma,i}^{(j)} Z_i^{(j)} \right\}, \qquad Z_i^{(j)} \overset{i.i.d.}{\sim} \mathcal{N}(0,1). \tag{1.6}$$

In other words, this specification represents small quote observation errors whose scale varies with moneyness and maturity, without imposing dependence across contracts. It serves as the low-noise benchmark case.

Figure 1 reports the resulting contaminated implied-volatility panels for a representative simulated date. In Design A, the perturbations are contract-specific and leave the broad shape of the clean surface intact. Figure 4 gives the corresponding realised perturbations for selected contracts over a 200-week subsample. The marginal scale variation underlying this design is shown in Figure 3.

1.3.1.2 Design B: Cross-sectionally correlated noise

A second observation error design allows for cross-sectional dependence in a given option panel for timepoint t_i by correlating errors across nearby strikes and tenors. This is largely motivated by the evidence and testing framework in Andersen et al. (2021), which explicitly considers spatial dependence in option observation errors. In this specification, the information content of a panel with q_i option prices is essentially not equivalent to that of q_i independent observations.

Let $\varepsilon_i^B = \left(\varepsilon_i^{(1),B}, \ldots, \varepsilon_i^{(q_i),B}\right)^\top$ and set

$$\tilde{\sigma}_i^{(j),B} = \max\left\{\sigma_{\min}, \sigma_i^{(j),0} + \varepsilon_i^{(j),B}\right\}, \qquad j = 1, \ldots, q_i, \tag{1.7}$$

where

$$\varepsilon_i^B \sim \mathcal{N}(0, D_i R_i D_i), \tag{1.8}$$

with

$$D_i = \operatorname{diag}\left(s_{\sigma,i}^{(1)}, \ldots, s_{\sigma,i}^{(q_i)}\right).$$

The correlation matrix R_i is given by

$$R_{i,jk} = \exp\left(-\frac{|m_i^{(j)} - m_i^{(k)}|}{\ell_m} - \frac{|\tau_i^{(j)} - \tau_i^{(k)}|}{\ell_\tau}\right). \tag{1.9}$$

This specification captures the possibility that nearby strikes and maturities share common quote or surface construction errors.

The cross-sectionally correlated design creates locally coherent distortions across nearby moneyness and maturity points. This is visible both in the contaminated panel in Figure 1 and in the signed perturbation plots in Figure 2.

1.3.1.3 Design C: Temporally persistent noise

The third observation error design allows for a persistent common component in the observation error, affecting the level, moneyness slope and maturity slope of the contaminated implied-volatility panel. This is consistent with the practical treatment of option panels as surfaces and with the broader evidence that option price noise can be quantitatively relevant for inference on risk premia as in Duarte et al. (2024).

Let $f_i = (f_{0,i}, f_{1,i}, f_{2,i})^\top$ evolve according to

$$f_i = A f_{i-1} + \xi_i, \qquad \xi_i \sim \mathcal{N}(0, Q), \tag{1.10}$$

where A is diagonal with entries inside the unit circle. For contract j, define

$$b_i^{(j)} = \left(1, m_i^{(j)}, \tau_i^{(j)}\right)^\top .$$

The raw implied-volatility observation error is

$$\tilde{\sigma}_i^{(j),C} = \max\left\{\sigma_{\min}, \sigma_i^{(j),0} + \left(b_i^{(j)}\right)^\top f_i + u_i^{(j)}\right\}, \qquad u_i^{(j)} \sim \mathcal{N}\left(0, \left(s_{\sigma,u,i}^{(j)}\right)^2\right), \quad (1.11)$$

where $s_{\sigma,u,i}^{(j)}$ is allowed to depend on $m_i^{(j)}$ and $\tau_i^{(j)}$.

The matrix A is the persistence matrix of the common observation-error factors. In the present specification it is taken to be diagonal, so that each component of f_i follows a separate first-order autoregressive process. The restriction that the diagonal entries of A lie inside the unit circle ensures stationarity. The matrix Q is the covariance matrix of the innovations ξ_i and controls the magnitude and possible co-movement of new shocks to the common components of the observation error.

The factor representation is a parsimonious way of introducing temporal dependence in the option panel observation errors. The vector f_i describes the common observation-error state at date t_i, while the loading vector $b_i^{(j)}$ determines how this common component affects option contract j, i.e., $\left(b_i^{(j)}\right)^\top f_i = f_{0,i} + m_i^{(j)} f_{1,i} + \tau_i^{(j)} f_{2,i}$. The first component acts as a common level error, the second component as a moneyness-slope error, and the third component as a maturity-slope error. This type of low-dimensional factor representation is consistent with the empirical treatment of implied-volatility panels as objects whose movements can be summarised by a small number of common components; see, for example, Cont and da Fonseca (2002) and related PCA-based evidence for implied-volatility surface such as Avellaneda et al. (2020).

Unlike Designs A and B, the persistent-factor design induces serially dependent quote distortions. Figure 5 reports the realised common factors and their effect on the same representative contracts over the same 200-week subsample as Figure 4.

For all three specifications, the raw contaminated price is obtained from the Black-Scholes pricing function:

$$\tilde{p}_i^{(j)} = B^{\mathrm{BS}}\left(S_i, K_i^{(j)}, \tau_i^{(j)}, r_i, q_i, \tilde{\sigma}_i^{(j)}\right). \quad (1.12)$$

The raw contaminated price $\tilde{p}_i^{(j),}$ is then rounded to the relevant tick size and projected back into the no-arbitrage price interval for the corresponding European option. The observed implied volatility $\sigma_i^{(j),\mathrm{obs}}$ is obtained by inverting the Black-Scholes pricing formula at the resulting observed price $p_i^{(j),\mathrm{obs}}$. This final step ensures that the contaminated panel contains both observed prices and observed implied volatilities, and that invalid prices generated by the noise mechanism are handled by capping rather than by re-sampling.

The detailed calibration of the noise parameters and the contract design $\mathcal{C}_i$ are deferred to Appendix B.

Note that the three observation error specifications are written in terms of model prices and Black-Scholes implied volatilities. Black-Scholes vegas are stored alongside the generated panels so that the same contaminated panels can also be used in vega-scaled estimation exercises if required.

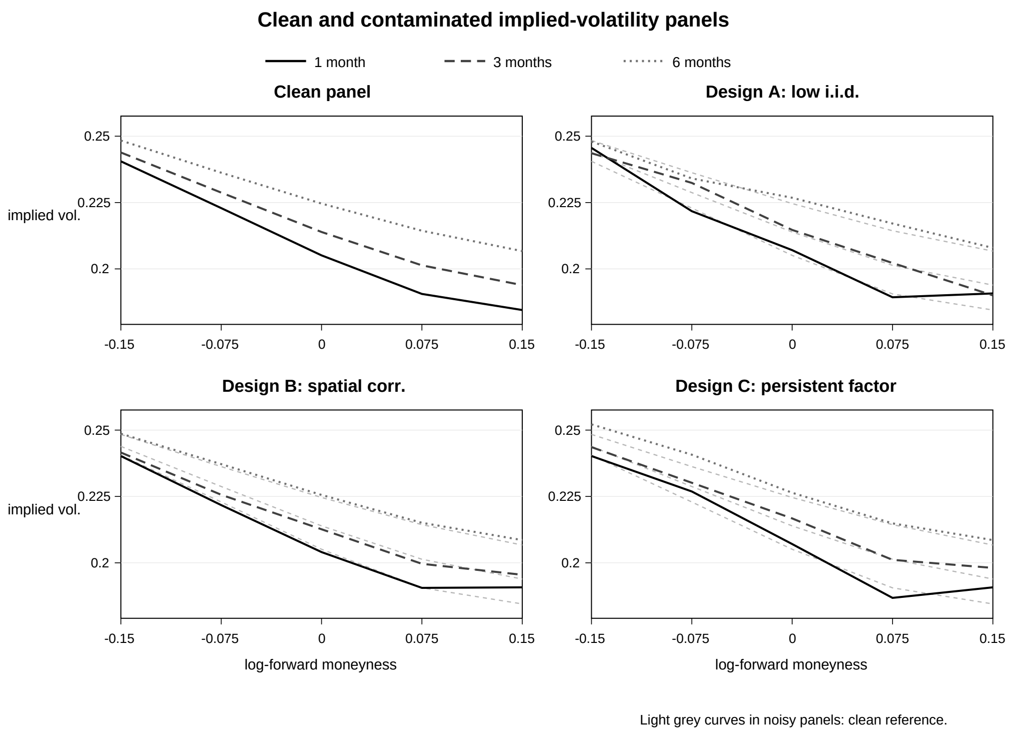

Figure 1: Clean and contaminated implied-volatility panels for sample 000 at week index 129. The clean panel is generated exactly from the Heston pricing model. Designs A–C introduce increasingly structured quote distortions in implied-volatility space before prices are reconstructed and projected into admissible price intervals.

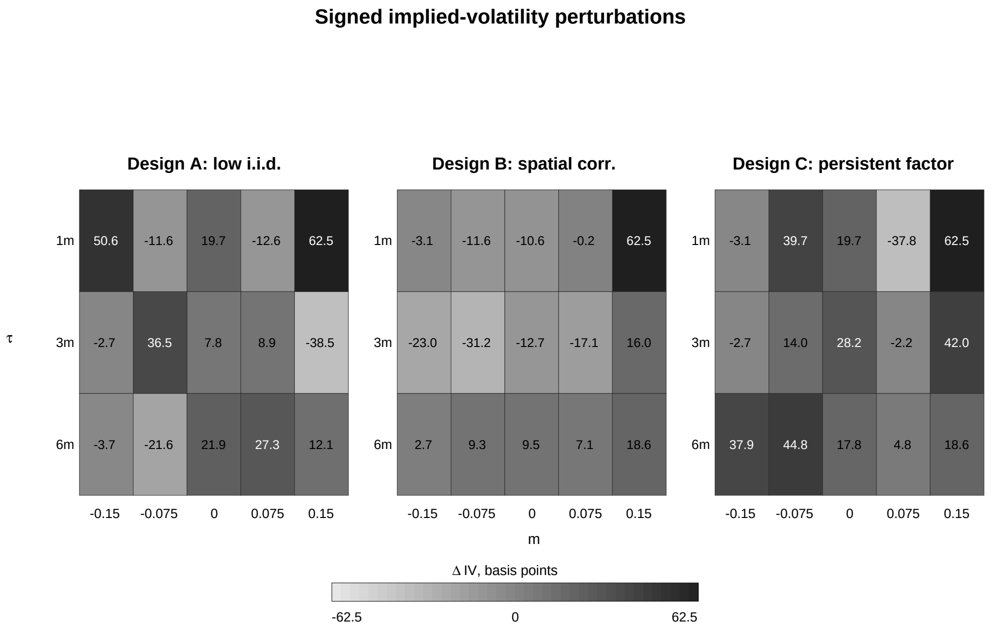

Figure 2: Signed implied-volatility perturbations relative to the clean model-implied panel for Designs A–C at week index 129. The plots show observed minus clean implied volatility across log-forward moneyness and maturity.

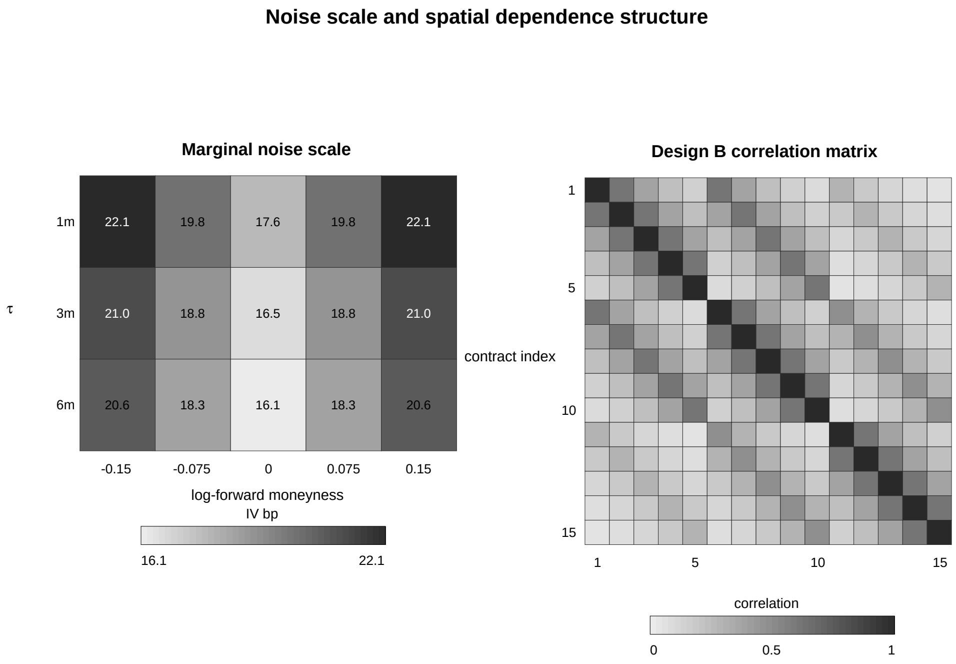

Figure 3: Cross-sectional structure of the observation-error designs. The left panel shows the marginal implied-volatility error scale. The right panel shows the spatial correlation matrix used in Design B after ordering contracts by maturity and log-forward moneyness.

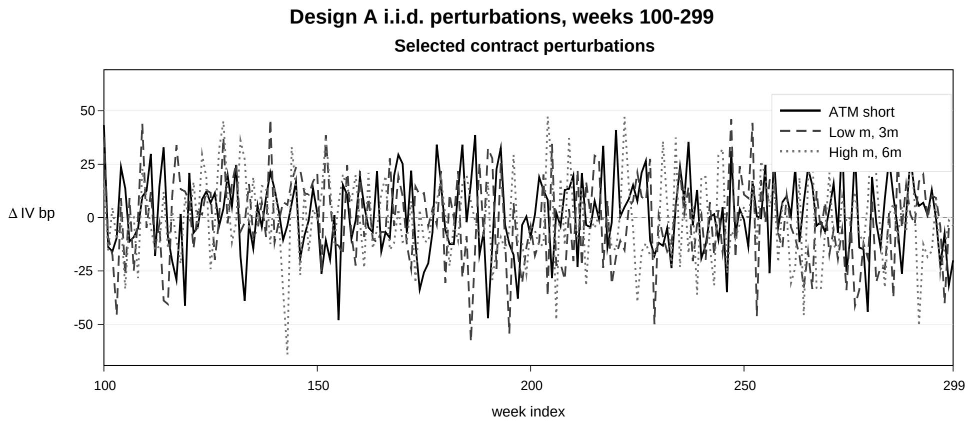

Figure 4: Realised perturbations for selected contracts under Design A over weeks 100–299. The series are independent across contracts conditional on the clean panel, so the plot contains no persistent common observation-error factor.

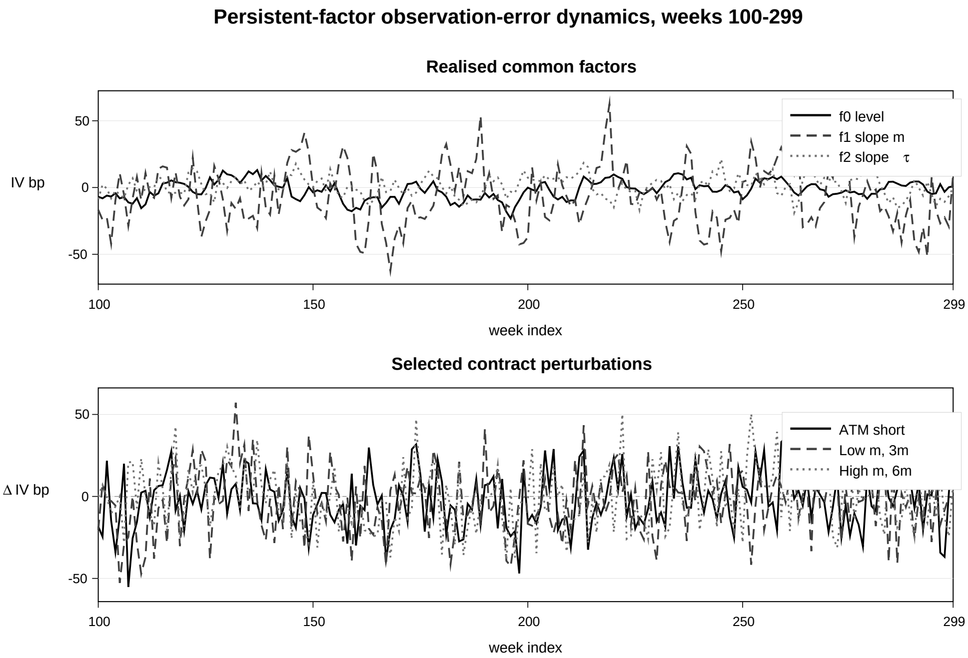

Figure 5: Temporal dependence in Design C over weeks 100–299. The upper panel shows the realised level, moneyness-slope and maturity-slope observation-error factors. The lower panel shows the resulting serially persistent implied-volatility perturbations for selected representative contracts.

1.3.2 Particle Filter Based Estimation

Estimation contexts which make use of option prices (that are nonlinear, often semi-analytic expressions featuring model parameters and partially observable state vectors) can be computationally demanding. The models used to describe index returns for option pricing purposes typically feature latent stochastic state processes which require filtering or implying. This increases the computational cost of econometric estimation exercises as one has to devise parsimonious estimators which can work with non-analytic expressions of state vector observations by relying on stable and robust numerical procedures.

Two other hurdles are typically encountered when working with continuous time models. Firstly, even the simple continuous time stochastic model specifications (like the base-case model used in this paper) do not have known densities readily usable to evaluate the likelihood function. Secondly, one often has to resort to using discretisation schemes to simulate state vector (i.e., particle) paths. However, despite this second hurdle, approximating transition densities using discrete distributions consisting of particle support points provides a tractable workaround which can easily be adapted for most of the typical model specifications considered. The filtering distributions provide estimators for the latent states and their possible distributions. The Monte Carlo procedures involved in the simulation of particles can be tailored to accommodate models with jumps and with more complex state dynamics than featured in the Heston model specification.

In the context of the base model specification previously introduced, we outline the following optimal filtering procedure in the spirit of Johannes et al. (2009), Christoffersen et al. (2010) and ***[Hurn et al]*** and introduce the necessary implementation details for the Monte Carlo estimation exercise. First, let us denote the log-likelihood function of the model under the physical probability measure by

$$\log \mathcal{L}_T = \log f(X_0) + \sum_{i=0}^{T} \log f(X_{i+1}|X_i; \theta^{\mathbb{P},\mathbb{P}\cap\mathbb{Q}}). \tag{1.13}$$

The evaluation of this likelihood function is unfeasible as the state vector is not fully observed. Also note that it does not provide any information about $\theta^{\mathbb{Q}}$, the parameters which feature in the dynamics of the model under the risk neutral probability measure.

To take into account option prices and to allow for the likelihood to be evaluated, a filtering rule using the time series observations for the components of the state vector which are observable is introduced alongside an option price panel likelihood observation in order to extract information about the latent states which are not observable. Let us denote by $\mathcal{O}_i$ the filtration containing the collection of sigma algebras generated by the time series of observable states and associated option prices. E.g., for our base model specification this is the filtration generated by $S_1, \ldots, S_T$ and the panel $p_1(T,K), \ldots p_T(T,K)$ of option prices for all available option maturities T and strike prices K. The level of the stochastic volatility process V_i is unobservable. To evaluate the likelihood contribution, the following need to be evaluated:

`Step 1.` T_{i+1} `likelihood contribution:`

$$\begin{aligned} f(X_{i+1}|\mathcal{O}_i;\theta) &= \int_{\mathcal{D}_V} f(S_{i+1}|V_{i+1},\mathcal{O}_i;\theta) f(V_{i+1}|\mathcal{O}_i;\theta)\,\mathrm{d}V_{i+1} \\ &= \int_{\mathcal{D}_V} f(S_{i+1}|V_{i+1},\mathcal{O}_i;\theta) \int_{\mathcal{D}_V} f(V_{i+1}|V_i,\mathcal{O}_i;\theta) \\ &\qquad\qquad \times f(V_i|\mathcal{O}_i;\theta)\,\mathrm{d}V_i\,\mathrm{d}V_{i+1} \end{aligned} \tag{1.14}$$

`Step 2. Latent state variable density update:`

$$f(V_{i+1}|\mathcal{O}_{i+1};\theta) = \frac{f(X_{i+1}, V_{i+1}|\mathcal{O}_{i+1};\theta)}{f(X_{i+1}|\mathcal{O}_{i+1};\theta)} \tag{1.15}$$

1.3.2.1 Current computational implementation.

At the current stage, the Heston simulation engine and the option-panel generation routines have been implemented in Python. The first-stage implied-state C-GMM code is also in place. It constructs scalar date-by-date implied-variance states from option-implied volatilities, evaluates a Heston $\mathbb{P}$-transition characteristic-function criterion, and forms a first-step C-GMM objective with analytically integrated Gaussian instruments. The transition characteristic function can be evaluated using either the analytic Heston Riccati solution or a Runge–Kutta reference solver. The results reported here do not include Monte Carlo estimation output; the present update concerns the simulation design, panel construction and visualisation of the observation-error mechanisms.

A Simulation of Heston Paths and Model Option Panels

This appendix documents the simulation procedure used to generate paths of the state vector $X_t = [\log(S_t), V_t]$ under the historical probability measure $\mathbb{P}$ and the associated panel of model-implied European option prices under the equivalent martingale measure $\mathbb{Q}$, based on the data-generating process (1.1)–(1.2) and pricing dynamics (1.3)–(1.4).

A.1 Sampling grid and contract set

Let $\{t_i\}_{i=0}^{T}$ denote an equally-spaced time grid with $t_{i+1} - t_i = \Delta > 0$. The simulated sample consists of $\{(\log(S_{t_i}), V_{t_i})\}_{i=0}^{T}$. At each date t_i, we define a (possibly time-varying) set of option contracts $\mathcal{C}_i = \{(T_i^{(j)}, K_i^{(j)}, \text{type}^{(j)})\}_{j=1}^{q_i}$ and compute a vector of model-implied prices $p_i^{\text{model}} \in \mathbb{R}^{q_i}$.

In the numerical design, 100 independent samples are generated. For each sample, the Heston state vector is simulated at a daily frequency and retained at a weekly frequency. The daily simulation step is set to

$$\delta = \frac{1}{252},$$

while weekly observations are obtained by retaining every fifth daily simulated value. The retained sample contains 525 weekly intervals, corresponding to 526 weekly state observations. A burn-in of 756 daily observations is used. The initial spot level is set to $S_0 = 100$, and the daily values of the simulated state vector are stored in addition to the weekly observations.

At each weekly observation date, the clean option panel contains options with maturities:

$$\tau \in \{\frac{1}{12}, \frac{1}{4}, \frac{1}{2}\}$$

and log-forward moneyness levels:

$$m \in \{-0.15, -0.075, 0, 0.075, 0.15\}.$$

The option type is selected using an out-of-the-money convention. Put options are used for negative log-forward moneyness values and call options are used for non-negative log-forward moneyness values. At-the-money options are treated as calls. Thus, each weekly panel contains 15 option contracts.

A.2 State simulation under $\mathbb{P}$

Let $y_t := \log(S_t)$ and write $y_i := y_{t_i}$ and $V_i := V_{t_i}$. Under $\mathbb{P}$, the dynamics are given by (1.1)–(1.2). We simulate V_{i+1} using the exact transition of the square-root process and then update y_{i+1} using a discretisation that preserves the leverage channel implied by ρ.

A.2.1 Exact transition for $V_{i+1} \mid V_i$

Conditional on V_i, the square-root process in (1.2) admits an exact transition representation in terms of a scaled noncentral chi-square random variable:

$$V_{i+1} \stackrel{d}{=} c\,\chi_{\nu}'^{2}(\lambda), \tag{A.1}$$

where

$$c = \frac{\sigma_v^2 \left(1 - e^{-\kappa\Delta}\right)}{4\kappa}, \qquad \nu = \frac{4\kappa\bar{V}}{\sigma_v^2}, \qquad \lambda = \frac{4\kappa e^{-\kappa\Delta}}{\sigma_v^2 \left(1 - e^{-\kappa\Delta}\right)} V_i. \tag{A.2}$$

This step guarantees $V_{i+1} \geq 0$ and removes Euler-type discretization bias in the variance marginal transition.

A.2.2 Integrated variance proxy

Define the integrated variance over $[t_i, t_{i+1}]$ by

$$I_i := \int_{t_i}^{t_{i+1}} V_s \,\mathrm{d}s. \tag{A.3}$$

Since I_i is not sampled explicitly, we use the trapezoidal proxy

$$\widehat{I}_i := \frac{\Delta}{2}\left(V_i + V_{i+1}\right). \tag{A.4}$$

A.2.3 Log-price update with leverage

Let $Z_{i+1} \sim \mathcal{N}(0,1)$ i.i.d. across i and independent of the variance draw. Using the decomposition of the Brownian drivers in (1.1)–(1.2), the log-price increment is simulated as

$$\Delta y_i := y_{i+1} - y_i$$
$$= (r_{t_i} - q_{t_i})\Delta + \left(\eta - \frac{1}{2}\right)\widehat{I}_i + \frac{\rho}{\sigma_v}\left[(V_{i+1} - V_i) - \kappa\left(\bar{V}\Delta - \widehat{I}_i\right)\right] + \sqrt{1-\rho^2}\sqrt{\widehat{I}_i}\, Z_{i+1}. \tag{A.5}$$

Finally, set $y_{i+1} = y_i + \Delta y_i$ and $S_{i+1} = \exp(y_{i+1})$.

A.3 Model option panels under $\mathbb{Q}$

Given the simulated state $X_i = [\log(S_i), V_i]$ at time t_i, model option prices are computed under $\mathbb{Q}$ for each contract $(T_i^{(j)}, K_i^{(j)}) \in \mathcal{C}_i$ by

$$p_i^{\text{model}}(T_i^{(j)}, K_i^{(j)}) = P\Big(X_i, T_i^{(j)}, K_i^{(j)}; \theta^{\mathbb{P}\cap\mathbb{Q}}, \theta^{\mathbb{Q}}\Big), \qquad j = 1, \ldots, q_i, \tag{A.6}$$

with $\theta = [\eta, \kappa, \bar{V}, \sigma_v, \rho, \eta_V]$ and the partitioning used in the main text. Under $\mathbb{Q}$, the variance drift (1.4) can be written in standard Heston form by defining

$$\kappa_{\mathbb{Q}} := \kappa - \eta_V, \qquad \bar{V}_{\mathbb{Q}} := \frac{\kappa\bar{V}}{\kappa - \eta_V}, \tag{A.7}$$

so that $dV_t = \kappa_{\mathbb{Q}}(\bar{V}_{\mathbb{Q}} - V_t)\, dt + \sigma_v\sqrt{V_t}\, dW_{t,v}^{\mathbb{Q}}$.

A.3.1 COS pricing

European option prices in (A.6) are evaluated using the Fourier-cosine (COS) method, which expands the risk-neutral density of the log-price (or log-return) on a bounded interval as a cosine series and integrates the payoff against this approximation.

Let $\tau := T - t_i$ and define the log-forward moneyness

$$x := \log\left(\frac{S_i e^{(r_{t_i} - q_{t_i})\tau}}{K}\right). \tag{A.8}$$

Denote by $\phi(u; \tau, X_i)$ the conditional characteristic function of $\log(S_T)$ under $\mathbb{Q}$ (which is available in semi-closed form for Heston). The COS approximation for a European call can be written as

$$C_i(T,K) \approx e^{-r_{t_i}\tau} \sum_{k=0}^{N-1}{}' \Re\left\{ \phi\left(\frac{k\pi}{b-a}; \tau, X_i\right) \exp\left(-i\frac{k\pi a}{b-a}\right) \right\} U_k(x), \tag{A.9}$$

where $[a,b]$ is a truncation range for the log-price (or log-return) support, N is the number of cosine terms, and $\sum'$ indicates that the $k=0$ term is weighted by $1/2$. The payoff coefficients $U_k(x)$ admit closed forms for vanilla calls and puts. Put prices are obtained either via the corresponding COS payoff coefficients or via put-call parity.

A.4 Parameter values used for clean sample generation

The historical-measure parameter values used in the data generating process are

$$\eta = 5.0, \qquad \kappa = 7.0, \qquad \bar{V} = 0.0225, \qquad \sigma_v = 0.4, \qquad \rho = -0.5,$$

with (r=0.02) and (q=0). The risk-neutral variance risk-premium parameter is set to

$$\eta_V = 5.0.$$

Given the notation in (1.4), this implies

$$\kappa_{\mathbb{Q}} = \kappa - \eta_V = 2.0, \qquad \bar{V}_{\mathbb{Q}} = \frac{\kappa \bar{V}}{\kappa - \eta_V} = 0.07875.$$

Clean option prices are computed using the COS method. The corresponding Black-Scholes implied volatilities are computed using the same implied-volatility inversion routine used later for contaminated panels.

B Option Panel Observation Error Specifications

This appendix documents the functional forms used to generate contaminated option panels from the clean model-implied option panel. The purpose is to introduce observation error structures which are simple enough to be implemented in a large-scale Monte Carlo design, but sufficiently close to observed option-market features to provide a meaningful stress-test of PF and IS estimation.

Noisy option panels are constructed from the clean model-implied panels generated in Appendix A. The clean panel at date t_i consists of model prices p_i^0, Black-Scholes implied volatilities σ_i^0, contract characteristics and Black-Scholes vegas. The observation error designs introduced in the main text are applied to the clean implied-volatility panel. The corresponding raw contaminated price is then obtained from the Black-Scholes pricing formula. Tick rounding, no-arbitrage projection and implied-volatility inversion are applied afterwards using the same procedure for all three designs.

B.1 Tick rounding, price validity and implied-volatility inversion

For each raw contaminated price $\tilde{p}_i^{(j)}$, tick rounding is applied through

$$\mathcal{R}_{\delta_{\text{tick}}}(x) = \delta_{\text{tick}} \text{round}\left(\frac{x}{\delta_{\text{tick}}}\right).$$

Let $L_i^{(j)}$ and $U_i^{(j)}$ denote the no-arbitrage lower and upper bounds for the corresponding European option. For a call option,

$$L_i^{(j)} = \max\left(S_i e^{-q\tau_i^{(j)}} - K_i^{(j)} e^{-r\tau_i^{(j)}}, 0\right), \qquad U_i^{(j)} = S_i e^{-q\tau_i^{(j)}}.$$

For a put option,

$$L_i^{(j)} = \max\left(K_i^{(j)} e^{-r\tau_i^{(j)}} - S_i e^{-q\tau_i^{(j)}}, 0\right), \qquad U_i^{(j)} = K_i^{(j)} e^{-r\tau_i^{(j)}}.$$

The observed price is obtained by projecting the rounded price into the admissible interval:

$$p_i^{(j),\text{obs}} = \min\left\{U_i^{(j)} - \epsilon_p, \max\left[L_i^{(j)} + \epsilon_p, \mathcal{R}_{\delta_{\text{tick}}}\left(\tilde{p}_i^{(j)}\right)\right]\right\}. \tag{B.1}$$

The projection in (B.1) is used instead of re-sampling. This keeps the distributional draws reproducible and preserves information about the frequency with which the noise mechanism generates boundary prices. The implementation therefore records whether capping is applied at the lower or upper bound.

The observed Black-Scholes implied volatility $\sigma_i^{(j),\text{obs}}$ is computed from $p_i^{(j),\text{obs}}$ using a robust implied-volatility inversion routine. In the implementation this is handled through a stand-alone implied-volatility solver interface so that the same routine can be used for clean panels, contaminated panels and later estimator diagnostics. The preferred numerical method is the rational approximation approach of Jäckel (2015), which is designed for fast and robust implied-volatility inversion.

B.2 Implementation of contaminated panels

The clean model-implied panels are generated once and stored before any observation error is applied. Each contaminated panel is then generated as a deterministic transformation of the clean panel, conditional on the noise specification and the random seed assigned to the sample and scenario. This avoids re-pricing the Heston model for each observation-error design.

The implementation records, for each contract, the clean model price, the clean Black-Scholes implied volatility, the raw contaminated price before validity enforcement, the rounded and capped observed price, the observed implied volatility, the noise scenario and indicators for whether lower- or upper-bound capping was applied. The capping policy is preferred to re-sampling because it preserves reproducibility and makes the frequency of invalid raw noise draws observable.

The same implied-volatility inversion routine is used for clean and contaminated panels. This is important because the subsequent IS and PF estimation exercises may use observed implied volatilities, price-space errors, or vega-scaled price errors. A single stand-alone implied-volatility solver therefore prevents the construction of different data sets from depending on different numerical inversion conventions.

B.3 Parameter Choices for the Observation Error Specifications

For Design A, the scale parameters are set to

$$\alpha_0 = 0.0015, \qquad \alpha_m = 2.0, \qquad \alpha_\tau = 0.05.$$

The value of α_0 fixes the baseline magnitude of the implied-volatility observation error. Expressed in decimal volatility units, $\alpha_0 = 0.0015$ corresponds to 15 implied-volatility basis points before the moneyness and maturity adjustments are applied. The parameters α_m and α_τ determine the relative variation of the error scale across the option panel. Given the log-moneyness grid used in the simulation design, $\alpha_m = 2.0$ implies that a contract with $|m| = 0.15$ has an implied-volatility observation error scale which is approximately 30 percent larger than that of an otherwise identical at-the-money contract. The maturity parameter $\alpha_\tau = 0.05$ implies a smaller adjustment which is largest for short-dated contracts. The maturity component is approximately 17 percent at $\tau = 1/12$, 10 percent at $\tau = 1/4$ and 7 percent at $\tau = 1/2$.

For Design B, the marginal scale parameters α_0, α_m and α_τ are kept equal to those used for Design A. The difference relative to Design A is therefore introduced through the dependence structure of the observation error, rather than through a different marginal error scale. The correlation length parameters are set to

$$\ell_m = 0.15, \qquad \ell_\tau = \frac{1}{3}.$$

Given the log-moneyness grid used in the simulation design, this implies that two adjacent moneyness points with the same maturity have an observation error correlation of approximately 0.61. The maturity length parameter implies correlations of approximately 0.61 between the one-month and three-month maturities, 0.47 between the three-month and six-month maturities, and 0.29 between the one-month and six-month maturities. The resulting correlation matrix therefore induces local dependence across the option panel, while avoiding an almost perfectly correlated surface error.

For Design C, the persistence matrix is set to

$$A = \mathrm{diag}(0.85, 0.65, 0.65).$$

The first diagonal element governs the persistence of the common level component of the observation error, while the second and third diagonal elements govern the persistence of the moneyness-slope and maturity-slope components. The level component is made more persistent than the slope components, reflecting the interpretation of this design as one in which part of the option panel observation error is carried across adjacent observation dates.

The innovation covariance matrix Q is specified through the stationary standard deviations of the three factor components. These are set to

$$(\bar{s}_{f,0}, \bar{s}_{f,m}, \bar{s}_{f,\tau}) = (0.0007,\ 0.0025,\ 0.0008).$$

Given $A = \mathrm{diag}(a_0, a_m, a_\tau)$, the innovation covariance matrix is

$$Q = \mathrm{diag}\left((1 - a_0^2)\bar{s}_{f,0}^2,\ (1 - a_m^2)\bar{s}_{f,m}^2,\ (1 - a_\tau^2)\bar{s}_{f,\tau}^2\right).$$

This gives

$$Q = \mathrm{diag}\left(1.35975 \cdot 10^{-7},\ 3.609375 \cdot 10^{-6},\ 3.696 \cdot 10^{-7}\right).$$

The residual idiosyncratic scale is chosen so that Design C has the same unconditional pointwise error scale as Design A. Let

$$\Omega_f = \operatorname{diag}\left(\bar{s}_{f,0}^2, \bar{s}_{f,m}^2, \bar{s}_{f,\tau}^2\right).$$

Then

$$s_{\sigma,u,i}^{(j)} = \sqrt{\max\left[\left(s_{\sigma,i}^{(j)}\right)^2 - \left(b_i^{(j)}\right)^\top \Omega_f b_i^{(j)}, 0\right]}.$$

Thus, the common component introduces temporal dependence and common variation across moneyness and maturity, while the total marginal implied-volatility observation error remains comparable to that of Designs A and B. This makes the comparison across the three designs depend mainly on the dependence structure of the observation error rather than on a mechanical increase in its absolute scale.

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